



Molecular diversity and polymerase gene genotypes of HIV-1 among treatment-naïve Cameroonian subjects with advanced disease

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ABSTRACT

Background: The progress of antiretroviral treatment roll-out programs in developing countries requires extensive monitoring of primary drug resistance prior to initiation of therapy. This is particularly relevant for Cameroon where a high HIV diversity has been reported.

Objectives: To determine HIV diversity in Yaoundé, Cameroon, in a cohort of HIV-infected subjects with advanced disease. To characterize HIV-1 mutations conferring primary drug resistance and to assess primary resistance patterns in the RNase H domain of the reverse transcriptase of these viruses.

Study design: HIV-1 RNA was extracted from plasma of 59 HIV-1 infected, drug-naïve subjects with CD4⁺ T-cell counts <200/μl. HIV-1 pol (PR, RT and RNase H) regions were sequenced for subtyping and for identifying drug resistance mutations in pol (PR, RT and RNase H).

Results: A complex HIV-1 diversity was seen, with multiple subtypes (A1, A2, C, D, F2, H, group O), CRFs (02_AG, 09_cpx, 11_cpx, 13_cpx, 22_01A1, 30_0206, 43_02G) and URFs. Primary drug resistance was low in PR (2%) and in RT regions (4%). RNase H mutations Q509L and Q547K were found in non-CRF02_AG strains.

Conclusions: A high HIV-1 diversity was already present in Cameroon in the early 90s, when the subjects were likely infected. Primary HIV-1 drug resistance was low. Occurrence of RNase H mutations with proven phenotypic effect on susceptibility to antiretrovirals encourages further assessment of their impact in treatment outcome in the context of complex HIV genetic diversity and in a subtype-specific fashion.

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1. Background

As international initiatives for the roll-out and expansion of antiretroviral therapy (ART) for HIV infection and AIDS progress in the developing world, monitoring HIV genetic diversity and pre-existence of circulating drug-resistant strains among drug-naïve individuals becomes increasingly essential. This is especially relevant for sub-Saharan countries, accounting for approximately three-fourths of worldwide infections and where ART still has a low coverage although with clear dissemination.¹ In Cameroon,

the HIV/AIDS official governmental program was launched in early 2000, and has become fully functional since 2005, with the support of the World Health Organization and the Global Fund.² HIV prevalence rates in Cameroon are around 5% overall and 10% in adults, although recent work has shown decreasing prevalence and regional heterogeneity.²

HIV-1 is divided into four groups, M, N, O and possible group P. Group M, responsible for the pandemic, can be divided into nine genetic subtypes and over 40 circulating recombinant forms (CRFs), mosaic variants derived from recombination between two or more subtypes.^{3,4} Cameroon is unique with respect to the diversity of circulating HIV strains, a country where all major types, groups, subtypes and CRFs have been reported.^{5–13} These studies have also evaluated the prevalence of mutations conferring drug resistance in treatment-naïve subjects, but most of them are restricted to recently infected subjects. Paradoxically, the current, still limited ART access programs in Cameroon are primarily directed to patients with advanced HIV disease who are in urgent need of treatment.

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2. Objectives

In this study, we evaluated the prevalence of HIV-1 groups, subtypes and prevalence of HIV-1 mutations conferring drug resistance in a cohort of late-stage HIV-infected subjects prior to their enrollment in a prospective ART study. We also assessed, for the first time, drug-associated mutations in the RNase H (RNH) domain of viral reverse transcriptase (RT) of treatment-naïve subjects in a setting of a high viral diversity.

3. Study design

Patients and samples. Fifty-nine patients with confirmed HIV infection and late-stage AIDS were recruited at the HIV/AIDS and Dermatology Clinic of the University of Yaoundé Teaching Hospital between 2002 and 2005. All patients were HIV-positive, ARV-naïve, had CD4 counts below 200 and signed an informed consent to participate in the study. Five milliliters of whole blood were collected from each subject, centrifuged, frozen at -70°C until viral subtyping and drug resistance testing were done.

RT-PCR and DNA sequencing. Extraction of HIV-1 RNA was done as previously described.^{14,15} Complementary DNA synthesis and PCR amplification of the HIV-1 *pol* region spanning the entire protease (PR) gene and approximately two-thirds of the reverse transcriptase (RT) gene (a 1152 bp fragment) were conducted in two steps with specific nested primers.¹⁵ Where this was not successful, PR and RT were separately amplified according to previously established protocols;¹⁵ in some cases only one of these fragments was obtained. The HIV-1 RNH region, corresponding to the C-terminal Pol fragment, was amplified for subtyping and analysis of nucleoside RT inhibitor-related mutations, as previously described.¹⁶ All PCR products were purified, and sequenced using the Big Dye v.3.1 kit (Applied Biosystems, Foster City, USA) with an automated ABI Prism ABI377 Genetic Analyzer (Applied Biosystems) according to the manufacturer's specifications and using inner PCR primers. Electropherograms were assembled with SeqMan (DNASTar, Madison, USA) and manually edited.

Virus subtyping and genotyping. HIV-1 subtyping was performed following sequence alignment (PR+RT, PR or RT) with reference sequences of all HIV-1 subtypes available in Los Alamos database (<http://hiv-web.lanl.gov>) using BioEdit Sequence Alignment Editor.¹⁷ Phylogenetic analyses were carried out with neighbor-joining (NJ) and Kimura's two-parameter model with MEGA 4.0¹⁸ and 1000 bootstrap replicates, using an alignment length of 1085 bp. Sequences which did not group with any subtype/CRF or with "outlier" behavior within clades, were further analyzed with Simplot v. 3.5.1 to determine recombination patterns.¹⁹ Selected, representative viral sequences placed within the "CRF02_AG-like" clade (CM04-081, CM04-098 and CM05-105) were also subjected to similarity plot analysis in Simplot to evidence their complex origin. PR and RT sequences were further analyzed for drug resistance mutations using the Stanford HIV Drug Resistance algorithm.²⁰ Mutations were identified according to the latest International AIDS Society–USA consensus²¹ and to the most recent WHO list of surveillance drug resistance mutations.²² RNH sequences of 29 strains were genotyped for identifying recently described drug exposure-associated mutations (K451R, D488E, I506L, Q547K and Q509L).^{16,23–25}

Sequences obtained in the study were submitted to the GenBank under accession numbers GU366105–GU366190.

4. Results

4.1. Genetic complexity of HIV-1 strains in Cameroon

Viral RNA was isolated from 59 HIV-positive patients, and in most strains (44; 75%) both PR and RT were sequenced, while PR

was the only region sequenced in 5 strains (8%) and RT in other 5. RNH sequences were generated from 29 strains, and in 5 patients it accounted for the only genomic fragment.

A high HIV genetic diversity of pure HIV-1 subtypes and CRFs was found: 31 CRF02_AG (51%), 5 CRF11_cpx (8%), 4 CRF22.01A1 (7%), 2 of each CRF09_cpx and D (3%), and 1 of each CRF43.02G, CRF30.0206, CRF13_cpx, A1, A2, C, F2, H and group O (2%). Five strains (10%) were assigned to unique recombinant forms (URFs) derived from different combination of subtypes.

Fig. 1 depicts an NJ tree grouping viral strains for which the largest *pol* fragments (PR+RT) were available. The predominant clade, of "CRF02_AG-like" strains, including intermingling clinical and reference strains of subtype G, CRF36_cpx and CRF30.0206, was not strongly supported (bootstrap < 70%). Moreover, most URF strains were placed in a sister clade respective to CRF02_AG strains, representing complex recombinants comprising subtypes A1, A2 and G and their derivatives CRF02_AG, CRF30.0206, CRF36_cpx and CRF22.01A1. Indeed, recombination analyses of those strains further corroborated these findings (data not shown). A similarity plot analysis of three "CRF02_AG-like" representative strains revealed complex evolutionary histories (Fig. 2). Strain CM04-081 showed a higher homology to CRF36_cpx reference strains throughout the analyzed fragment (Fig. 2A). CM04-098 showed unclear homology to CRF02_AG and CRF36_cpx strains in the 5' half of the fragment, and to CRF30.0206 in the 3' half (Fig. 2B). Finally, CM05-105 showed intermingling regions of homology with CRF36_cpx and CRF30.0206 (Fig. 2C). These results were good evidence of the complexity of HIV-1 strains in Cameroon.

Some strains depicted in Fig. 1 (CM04-089, CM04-093 and CM04-095) strongly grouped with reference strains of CRF22.01A1, a variant derived of CRF01_AE and subtype A1. We did not find any CRF01_AE strains in our survey despite recent descriptions of this HIV variant in Cameroon.^{10,11}

4.2. Drug resistance mutations and primary resistance

In PR, a high prevalence of secondary drug resistance mutations was found, characteristic of non-B subtypes (data not shown). Two strains harbored primary mutations; CM04-043 carried mutations D30N and M46I, while CM04-052 carried only M46I (Table 1). Three strains showed mutations of unknown significance to drug resistance at PR position 48, G48R in CM04-049 and CM04-053, and G48E in CM04-043. Finally, CM04-090 harbored a L33I mutation. In RT, one strain (CM04-089) showed a K219Q mutation. Additional mutations at RT codons associated with drug resistance were found in strains CM04-085 (V106I), CM05-103 (T69S) and CM05-101 (K103R). All mutations were found in CRF02_AG strains except for K103R, found in a CRF11_cpx virus. We did not find any additional mutation of the WHO surveillance drug resistance mutation list.²²

In the RNase H domain, we found mutations Q509L and Q547K in one CRF22.01A1 and in one group O strain, respectively (Table 1). Other mutations occurring at some of these sites were also observed, as was the case of mutations Q509H (one CRF13_cpx and one URF) and Q509I (one CRF11_cpx).

5. Discussion

In this study, an extensive genetic diversity of HIV-1 was found among Cameroonians with advanced stages of infection. These findings suggested that a high molecular HIV heterogeneity must have been already present in Cameroon in the early to mid 1990s, in agreement with previous reports.^{26,27} The observed genetic forms and prevalence were consistent with previous reports on molecular HIV diversity in this country.^{5–13}

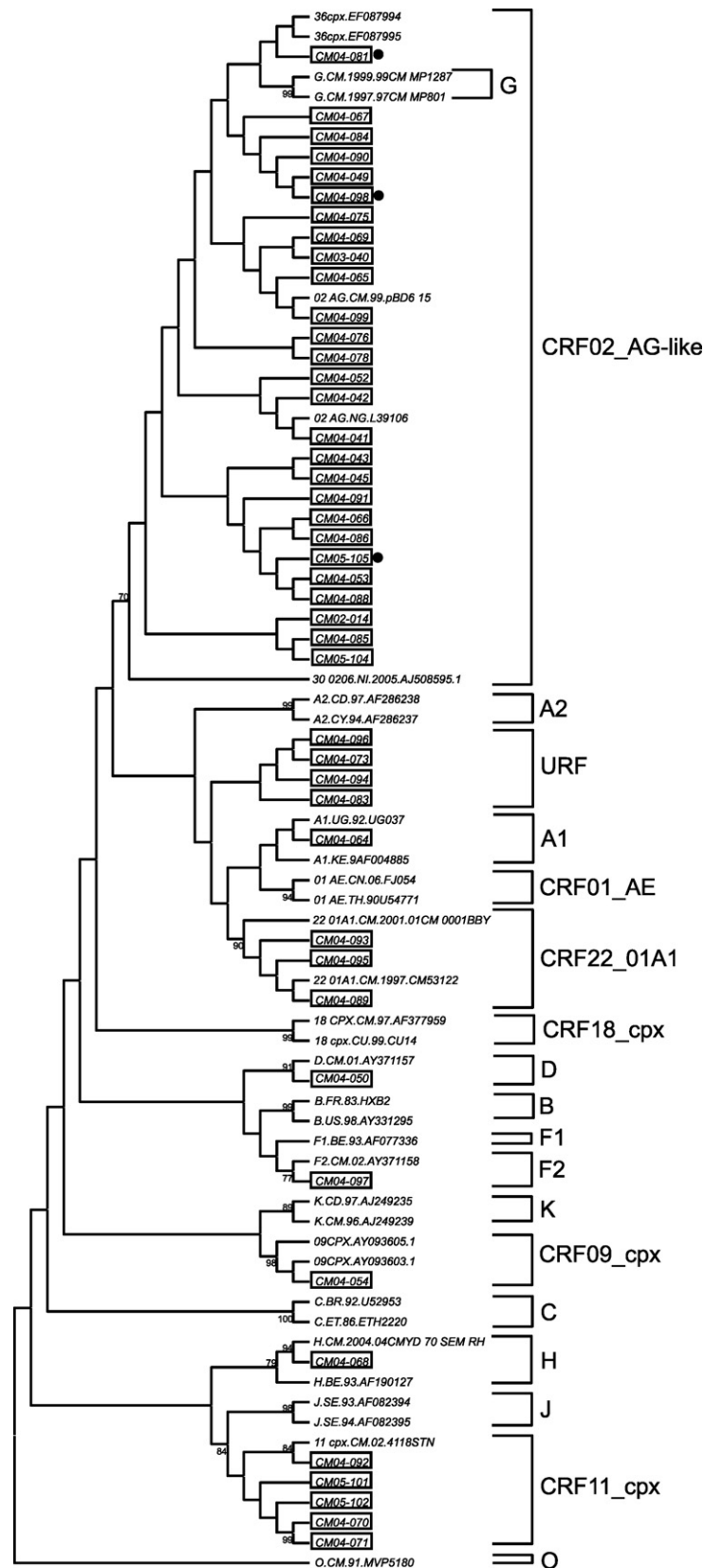


Fig. 1. Neighbor-joining tree for HIV-1 subtyping with representative HIV-1 sequences, including all strains for which the large *pol* fragment (PR and RT) was available. Nomenclature of clinical samples herein characterized is as follows: CM-yr of isolation–patient code. The tree includes reference sequences of the major HIV-1 subtypes and CRFs retrieved from the Los Alamos HIV Database with their original designations. Nomenclature is as follows: subtype–country–year–name of strain. Clinical samples characterized herein are boxed in the tree. The three samples subjected to diversity plot analysis (Fig. 2) are marked with black circles. Only significant bootstrap values ($\geq 70\%$) are shown. The analysis was based on a 1085 bp sequence alignment.

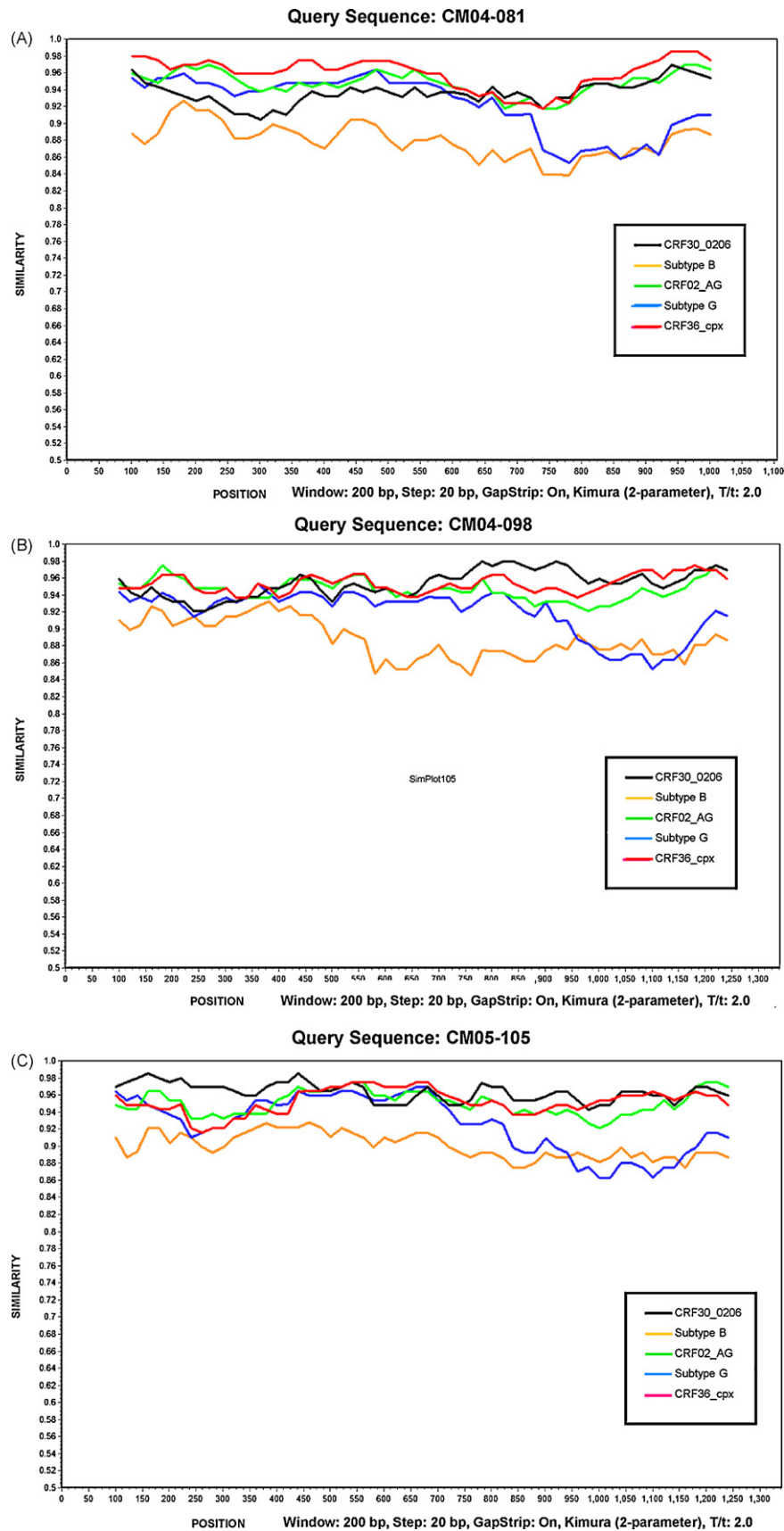


Fig. 2. Diversity plot analyses of representative complex recombinant HIV-1 viruses estimated with SimPlot v. 3.5.1. Query sequences were A, CM04-081; B, CM04-098; and C, CM05-105. Reference sequences and respective line colors for each comparison are depicted at the upper right inset of each panel. Parameters used in each analysis (window size, step size and evolutionary model) are listed below each graphic.

Table 1

Demographic, clinical and molecular data of patients from Yaoundé, Cameroon (2002–2005).

PATIENT CODE	Age	Sex	CD4 ^a	HIV CLADE	MUT PR	MUT RT	MUT RNH
CM02-014	–	–	–	CRF02_AG	Ø	Ø	–
CM02-016	34	M	41	CRF02_AG	–	Ø	–
CM03-035	35	F	117	CRF02_AG	–	–	Ø
CM03-040	47	M	147	CRF02_AG	Ø	Ø	–
CM04-041	37	F	9	CRF02_AG	Ø	Ø	Ø
CM04-042	60	M	76	CRF02_AG	Ø	Ø	Ø
CM04-043	41	M	11	CRF02_AG	D30N^a, M46I, G48E	Ø	Ø
CM04-045	27	F	39	CRF02_AG	Ø	Ø	Ø
CM04-049	53	F	4	CRF02_AG	G48R	Ø	–
CM04-052	36	F	–	CRF02_AG	M46I	Ø	–
CM04-053	57	F	145	CRF02_AG	G48R	Ø	Ø
CM04-065	41	M	177	CRF02_AG	Ø	Ø	–
CM04-066	22	F	12	CRF02_AG	Ø	Ø	–
CM04-067	30	F	25	CRF02_AG	Ø	Ø	–
CM04-069	42	F	–	CRF02_AG	Ø	Ø	Ø
CM04-072	36	F	161	CRF02_AG	–	Ø	–
CM04-075	48	M	147	CRF02_AG	Ø	Ø	–
CM04-076	28	M	102	CRF02_AG	Ø	Ø	Ø
CM04-078	36	F	29	CRF02_AG	Ø	Ø	Ø
CM04-081	53	F	67	CRF02_AG	Ø	Ø	Ø
CM04-084	30	M	1	CRF02_AG	Ø	Ø	–
CM04-085	49	F	110	CRF02_AG	Ø	V106I	–
CM04-086	29	F	14	CRF02_AG	Ø	Ø	Ø
CM04-088	33	F	24	CRF02_AG	Ø	Ø	Ø
CM04-090	26	F	8	CRF02_AG	L33I	Ø	Ø
CM04-091	26	F	1	CRF02_AG	Ø	Ø	Ø
CM04-098	38	F	65	CRF02_AG	Ø	Ø	Ø
CM04-099	23	F	9	CRF02_AG	Ø	Ø	Ø
CM05-103	26	F	102	CRF02_AG	–	T69S	–
CM05-104	36	F	123	CRF02_AG	Ø	Ø	–
CM05-105	42	F	17	CRF02_AG	Ø	Ø	Ø
CM03-029	32	F	41	URF	–	–	Ø
CM04-044	24	F	23	URF	–	Ø	Q509H
CM04-073	38	F	146	URF	Ø	Ø	–
CM04-083	58	M	25	URF	Ø	Ø	–
CM04-096	32	F	23	URF	Ø	Ø	–
CM04-094	49	M	17	URF	Ø	Ø	Ø
CM04-070	22	F	112	CRF11_cpx	Ø	Ø	–
CM04-071	27	M	159	CRF11_cpx	Ø	Ø	–
CM04-092	48	M	31	CRF11_cpx	Ø	Ø	Ø
CM05-101	42	F	135	CRF11_cpx	Ø	K103R	–
CM05-102	32	F	112	CRF11_cpx	Ø	Ø	Q509I
CM02-019	29	F	89	CRF22.01A1	Ø	–	–
CM04-089	53	M	13	CRF22.01A1	Ø	K219Q	Ø
CM04-093	29	F	147	CRF22.01A1	Ø	Ø	–
CM04-095	56	F	–	CRF22.01A1	Ø	Ø	Q509L
CM04-054	43	M	177	CRF09_cpx	Ø	Ø	–
CM04-077	37	M	175	CRF09_cpx	Ø	–	–
CM02-010	–	–	–	D	Ø	–	–
CM04-050	36	F	–	D	Ø	Ø	–
CM04-064	43	F	117	A1	Ø	Ø	–
CM03-032	28	M	47	A2	–	–	Ø
CM04-082	53	M	1	C	–	Ø	Ø
CM04-097	30	F	14	F2	Ø	Ø	–
CM04-068	60	F	148	H	Ø	Ø	Ø
CM04-051	38	M	172	CRF13_cpx	–	–	Q509H
CM04-055	38	M	78	CRF30-0206	Ø	–	–
CM05-106	38	F	26	CRF43-02G	Ø	–	–
CM04-100	46	M	7	Group O	–	–	Q547K

Ø, No mutations found. –, Unavailable data.

^a CD4⁺ T-cell counts at time of study enrolment.^a Mutations in bold refer to specific changes associated with drug resistance; the remaining correspond to other changes at codons associated with resistance.

Interestingly, a high genetic complexity of “CRF02_AG-like” strains was found in our survey. Although a phylogenetic reconstruction showed a “CRF02_AG-like” clade (Fig. 1), it was weakly supported by low bootstrap estimates and by the inclusion of several CRF02_AG-derived CRFs, like CRF30.0206,²⁸ CRF36cpx²⁹ and CRF43.02G.³⁰ Analysis of several, recently reported Cameroonian strains assigned to CRF02_AG showed that they rather represented strains of the abovementioned variants (data not shown). Similarly, we did not find any CRF01_AE strain, although three of our strains

strongly grouped with CRF22.01A1, a CRF comprising fragments of CRF01_AE. More meticulous analyses of Cameroonian HIV-1 strains should be conducted in further molecular epidemiology studies for allowing a more precise appreciation of HIV-1 genetic diversity.

In our survey for primary drug resistance, we did not find mutations conferring resistance to non-nucleoside RT inhibitors (NNRTI), reflecting lack of disseminated NNRTI usage in Cameroon at the time of collection. Our primary drug resistance estimates were in agreement with other recent surveys conducted in

Cameroon,^{9,31,32} except for the higher NNRTI resistance estimates observed in these reports.

New drug-associated mutations have been recently described in the C-terminal domains of HIV-1 RT, namely the connection and the RNH domains.^{16,23–25} While most studies have been conducted with HIV-1 subtype B, we have reported, for the first time, the occurrence of mutations in the RNH domain of drug-naïve viruses belonging to non-B subtypes. Analysis of 29 RNH sequences showed Q509L in one CRF22_01A1 strain and Q547K in an HIV-1 group O strain. The Q509L mutation has been shown to increase AZT resistance of TAM-containing viruses up to 50-fold when combined with the connection mutation A371V.²³ It has been suggested that Q509L increases AZT-monophosphate excision displayed by resistant HIV-1 RT.³³ The Q547K mutation has also been associated with thymidine analogue drug exposure.¹⁶ Although its mechanism of action has not been elucidated, Q547K lies within the RNH catalytic site and may reduce RNH activity, a phenomenon associated with increased resistance to thymidine analogues.^{24,34} Analysis of the natural occurrence of Q509L among drug-naïve CRF22_01A1 strain is not currently feasible due to the scarce number of these variants in public databases. However, a survey at the Stanford HIV Drug Resistance Database²⁰ conducted for the most common HIV-1 subtypes and CRF failed to show significant prevalence (<1%) of this mutation in any subtype. With respect to Q509H, found in an URF and in a CRF13_cpx, the Stanford database showed increased prevalence (13%) in CRF02_AG strains exposed to RT inhibitors when compared to drug-naïve strains, which may indicate that this mutation is also associated to transmitted drug resistance. The possibility that Q509H represents a polymorphism in certain HIV-1 non-B subtypes, however, cannot be ruled out.

Our analysis revealed that 30 of 32 RNH sequences carried Q547K among HIV-1 group O strains analyzed in the Los Alamos HIV Database. Although detailed information on drug exposure of these strains was not available, we presume that HIV-1 group O strains are likely carriers of this polymorphism with a high frequency. It remains to be determined whether such polymorphism has a phenotypic impact on drug susceptibility to RT inhibitors, either *per se* or in combination with primary resistance mutations.

In summary, we have shown a highly complex molecular diversity of HIV-1 strains in Cameroon in infected patients at an advanced stage of disease. This complexity makes HIV subtyping a difficult task in this geographic area, which should be conducted with caution. Moreover, we showed that levels of HIV-1 primary drug resistance were low in these patients. Finally, we were able to pinpoint newly described RNH drug-associated mutations, setting the ground for defining primary resistance in this RT C-terminal genomic region. Altogether, these data contribute to a better understanding on the impact of HIV-1 genetic diversity in primary resistance and in potential outcomes of ART in a highly diverse, resource-limited setting.

Conflicts of interest

None declared.

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